MIL-R-6130C
18 July 1980
SUPERSEDING
MIL-R-6130B
27 August 1963

MILITARY SPECIFICATION

RUBBER, CELLULAR, CHEMICALLY BLOWN

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification establishes requirements for chemically blown cellular rubber

1.2 Classification. The cellular rubber shall be furnished in the following types, grades and conditions:

1.2.1 Types.

Type I - Open cell (sponge rubber)
Type II - Closed cell (expanded rubber)

1 2.2 Grades.

Grade A - Oil and flame resistant
Grade B - Nonoil resistant
Grade C - Low temperature, nonoil resistant

1.2.3 Conditions.

Soft
Medium
Firm

2. APPLICABLE DOCUMENTS

2 1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 9321), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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SPECIFICATIONS

FEDERAL

C-C-91	-Candle, Illuminating
PPP-B-576	-Box, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-591	-Box, Fiberboard, Wood-Cleated
PPP-B-601	-Box, Wood; Cleated Plywood
PPP-B-621	-Box, Wood, Nailed and Lock-Corner
PPP-B-636	-Box, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-Marking for Shipment and Storage
MIL-STD-293	-Visual Inspection Guide for Cellular Rubber Items

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2 2 Other publications The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

ASTM D 297-79	-Rubber Products-Chemical Analysis
ASTM D 471-79	-Rubber Property-Effect of Liquids
ASTM D 1055-69 (Reapproved 1975)	-Flexible Cellular Materials-Latex Foam
ASTM D 1056-78	-Flexible Cellular Materials-Sponge or Expanded Rubber

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103)

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Uniform Classification Committee, Agent

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials. The materials shall be homogeneous. Except for the production allowances specified in 3 2.4, the material shall not include cemented, bouded, shredded, or reprocessed cellular rubber Grade A cellular rubber shall not contain natural rubber. Grades B and C cellular rubber shall be unrestricted in composition.

3.2 Design and construction

3.2.1 Form. Chemically blown cellular rubber shall be furnished as sheets or molded shapes as specified, and shall have a vulcanized uniform cellular structure, with thin skin or rind surfaces. Cut or split sheets void of skin or rind surfaces may be furnished only for Type II material when specified by the procuring activity (see 6 2).

3.2.2 Skin or rind The smooth surface formed by contact with the mold shall be considered a skin or rind It shall be of the same compound, and vulcanized integrally with the cellular structure

3.2.2.1 Cut or split sheets (Type II only). When two or more sheets are derived from one thick sheet, the cut or split sheets shall not be required to possess a skin or rind on either major surface (see 3 2 1). Only longitudinal splitting of the thickness shall be permitted.

3.2.3 Surfacing. Cellular rubber shall be backed or surfaced with fabric, cement, or other materials when and as specified in the contract or order (see 6.2), or by applicable drawings

3 2.4 Production allowances. The following defects and manufacturing methods are permitted in chemically blown cellular rubber when their occurrence is incident to standard manufacturing practice and assembly, provided they are corrected as indicated herein and the resulting product conforms with the requirements of this specification and applicable supplementary specifications and drawings:

3.2.4.1 Defects.

a. Tears occurring during removal from mold, carefully cemented together.

b. Changes in the volume of product during cure resulting in shallow surface depressions or minor changes in size (not exceeding 1 inch) corrected by addition of material of identical composition and quality as that of the product, and buffing to a level, neat-appearing surface

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3 2 4 2 Manufacturing methods

a Assembled products, where the finished product desired is larger than it is standard practice in industry to mold in one section

b Hand-built shapes, where the order is small and the nature of the shape is such as to make molding uneconomical

3.3 Finish

3.3.1 Sheets Sheets shall have a smooth skin or rind on both faces, unless otherwise specified (see 3.2.1) Fabric impressions are not objectionable

3.3.2 Molded shapes. Molded shapes shall have a smooth skin or rind on all surfaces, unless otherwise specified. Marks left by mold partings or gates should be as small as practicable.

3.4 Color. The color of the material shall be "as manufactured," unless otherwise specified.

3.5 Odor The cellular rubber shall be free from any objectionable odor When specified, material shall be treated to minimize the odor (see 6 2).

3 6 Physical properties Physical properties of the cellular rubber shall conform to the requirements specified in Table I.

TABLE I. Physical properties.

Property	Requirement 1/		Test paragraph
	Type I	Type II	
Apparent density lbs/cu in. (kg/m^3) max			4 5 1
soft	0.020 (0.554 x 10^3)	.015 (0 415 x 10^3)	
medium	0.027 (0.747 x 10^3)	.020 (0.554 x 10^3)	
firm	0 030 (0.830 x 10^3)	030 (0.747 x 10^3)	
Compression deflection load, psi (kPa), for 25% deflection			4.5.2
soft	2 to 7 (14 to 48)	2 to 7 (14 to 48)	
medium	over 7 to 14 (48 to 96)	over 7 to 14 (48 to 96)	
firm	over 14 to 21 (96 to 144)	over 14 to 21 (96 to 144)	

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TABLE I. Physical properties. (continued)

Property	Requirement 1/		Test paragraph
	Type I	Type II	
Low Temperature resistance, % retention of orig. 25% deflection			
Grade A	30	30	
Grade B	30	30	
Grade C	45	45	
Accelerated aging, % retention of original 25% deflection	55 to 145	70 to 130	4.5.4
Recovery %, min after 50% deflection	90	70	4.5.5
Flexing, % set, max	10	10	4.5.6
Flame resistance, seconds, max, Grade A, only	30	30	4.5.7
Shrinkage, %, max	-	15	4.5.8
Water absorption, %, max	-	5	4.5.9
Oil aging, % change in volume			4.5.10
Grade A, only	-15 to +30	-15 to +30	
Sulfur content, %, by wt, max.	0.60	0.60	4.5.11

1/ Unless otherwise specified, the requirement is for all types, grades and conditions.

3 7 Dimensions and tolerances.

3.7 1 Dimensions. The dimensions of sheets or molded shapes shall be as specified by the procuring activity (see 6.2).

3.7 2 Tolerances.

3 7.2.1 Sheets. Unless otherwise specified, the tolerance requirements for thickness, lengths and width of sheets shall be as specified in Table II

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TABLE II. Dimensions and tolerances.

Dimension	Tolerance	
Thickness	Type I	Type II
under 0.125 inch (3.18 mm)	±0.016 (0 4 mm)	+0.047 (1.19 mm) -0.016 (0.4 mm)
0 125 to 0.500 (3.18 to 12.7 mm)	±0.031 (0 8 mm)	±0.063 (1.59 mm)
over 0.500 (12 7 mm)	±0.063 (1 59 mm)	±0.094 (2.38 mm)
Length & width		
under 36 inch (91 cm)	±0.25 (6.35 mm)	±0.25 (6.35 mm)
36 (91 cm) and over	+0.75 (1.91 cm) -0.0	+0.75 (1.91 cm) -0.0

3 7 2 2 Molded shapes. Tolerance requirements for molded shapes shall be as specified in drawings, contracts or by the procuring activity (see 6 2)

3 8 Identification marking Each piece of cellular rubber shall be marked, if size permits, in a permanent and legible manner using marking material which will not harm the rubber. The marking shall include.

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Type
Grade
Condition
NSN (when applicable)
Manufacturer's part no. (when applicable)
Contract or order no.
Manufacturer's name or trade mark
Quarter and calendar year of cure

3 8 1 Sheets. Marking on sheet material shall be applied to one side only in rows spaced not more than 5 inches (12.7 cm) apart in constantly recurring characters not less than 0.375 inch (9.53 mm) in height extending either lengthwise or crosswise over the sheet.

3.9 Age controls. Cellular rubber covered by this specification shall not be delivered to any government activity when it is more than 4 quarters old (see 3.8).

3 10 Workmanship. The chemically blown cellular rubber shall be manufactured by such processes as to produce materials meeting all the requirements of this specification.

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified

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in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The examination and testing of the cellular rubber shall be classified as quality conformance inspection (see 4.3).

4.3 Quality conformance.

4.3.1 Lot formation. Unless otherwise specified, a lot shall consist of all the material of the same type, grade and condition, manufactured at one time, forming part of one contract or order and submitted for acceptance at the same time and place.

4.3.2 Prior approval. Whenever a manufacturer has delivered an acceptable product meeting all the requirements of this specification, the procuring activity may, at its discretion, waive the inspections so identified by Footnote 2/ in Table V for a period of time not to exceed two years. This option applies only to the products of the same type, grade and condition.

4.3.3 Sampling and inspection procedures.

4.3.3.1 Visual. The sample unit for visual inspection shall be one sheet or molded part as applicable. Samples shall be randomly selected from each lot in accordance with Inspection Level I of MIL-STD-105. Each sample unit shall be inspected as specified in Table III. Odor examination shall be conducted at this time and be included as part of the defect scoring. The Acceptable Quality Level (AQL) shall be 2.5 major and 6.5 total defects per 100 units.

TABLE III. Visual inspection.

Examination	Defect	Major	Minor
Appearance	Not uniform in texture, finish or firmness.	X	
	Dirt, foreign material, imbedded particles.		X
	Production defects not corrected (3.2.4).	X	
Construction	Any defects classified as major or minor as specified in MIL-STD-293.		X
	Skin or rind not as specified.		X
Odor	Objectionable.		X

4.3.3.2 Dimensional. The sample unit shall be one sheet or molded part as applicable. The sample shall be selected in accordance with Inspection Level S-3 of MIL-STD-105, and may be randomly selected from other

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samples visually inspected. The sample shall be inspected in accordance with Table IV. The AQL shall be 1.5 percent defective.

TABLE IV. Dimensional inspection.

Examination	Defect
Sheets and molded parts	Any dimension that varies more than the plus or minus tolerances specified in 3.7, the applicable drawing or the contract

4.3 3.3 Physical properties. Sufficient cellular rubber shall be selected from each lot to complete the testing specified in Table V. Failure of the product to pass any requirement in Table V shall be cause to reject the lot represented by the material.

TABLE V. Physical properties.

Property	Rqmt para	Test para	No. of specimens	Results reported to 1/
Apparent density	Table I	4.5 1	3	0 001 unit
Compression deflection	Table I	4.5 2	3	0.1 psi
2/ Low temperature resistance	Table I	4.5.3	3	0.1%
2/ Accelerated aging (heat)	Table I	4 5.4	3	1%
Recovery	Table I	4.5.5	3	0.1%
2/ Flexing	Table I	4.5 6	3	0.1%
2/ Flame resistance	Table I	4.5.7	2	Second
2/ Shrinkage	Table I	4 5.8	3	0.1%
Water absorption, Type II	Table I	4.5.9	3	0.1%
2/ Oil aging	Table I	4.5.10	2	0 1%
2/ Free sulfer	Table I	4.5.11	3	0.01%

1/ All test results shall be reported as the average value All results shall be shown on test report.
2/ These inspections may be waived by the procuring activity in accordance with the provisions of 4.3.2.

4 3.3.4 Packaging inspection. Shipping containers, just prior to closure shall be randomly selected from each lot in accordance with Inspection Level I of MIL-STD-105 The sample unit shall be one shipping container. The lot size shall be the number of shipping containers. The AQL shall be 4.0 percent defective. Samples shall comply with the requirements of Section 5 and Table VI In addition, shipping containers fully prepared for delivery shall be examined for closure defects.

TABLE VI. Packaging inspection

Examine	Defect
Packaging (as applicable) Sheets	Paper separator sheets omitted

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TABLE V. Packaging inspection.

Examine	Defect
Molded shapes	Not wrapped, boxed or otherwise protected against abrasion and deformation. Unit package not as specified. Packaging material not as specified.
Packing	Not in accordance with contract requirements. Arrangements or number of units per shipping container not in accordance with contract requirements. Container not as specified or required methods or materials.
	Any non-conforming component, component missing, damaged or otherwise defective affecting serviceability
	Inadequate application of components, such as: incomplete closures of case liners, container flaps, loose or inadequate strapping, bulged or distorted containers.
Count	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Interior or exterior markings (as applicable) omitted, illegible, incorrect, incomplete or not in accordance with contract requirements. Precautionary markings missing or not as specified.

4 4 Testing conditions.

4 4 1 Standard conditions. Unless otherwise specified herein, all tests shall be conducted at 24° ±3°C (75° ±5°F) and a relative humidity of 50 ±4 percent.

4 4 2 Specimen conditioning. Unless otherwise specified, all test specimens shall be conditioned at standard conditions for at least 22 hours prior to testing.

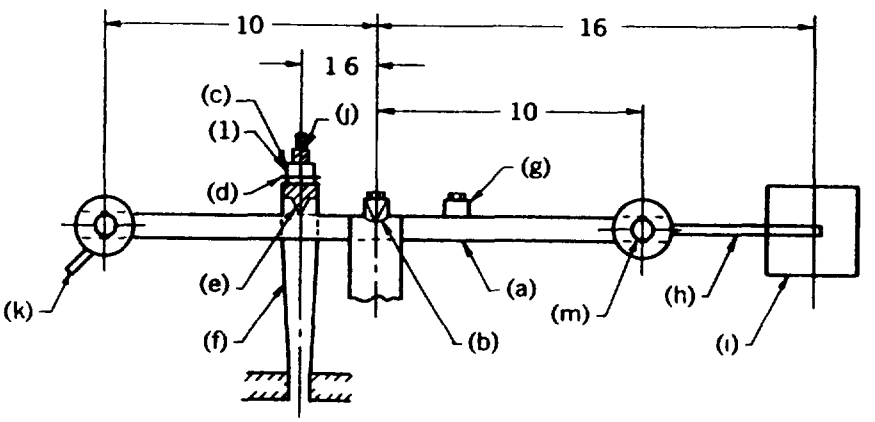
4.5 Test methods.

4.5 1 Apparent density. The density of regular shaped items (easily measured parts and sheets) shall be determined by accurately measuring to determine volume and weighing. Three samples shall be used. The density of irregular shaped items shall be determined by any acceptable procedure. The free flowing powder or seed method is one such procedure.

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4.5.2 Compression deflection.

4 5.2.1 Apparatus The test apparatus shall be such that the deflection plates remain parallel when the load is applied. Apparatus as shown in Figure 1 may be used



Deflection lever ratio of machine is 16 to 1.6

Deflection of 0.10 inch at specimen equals a vertical displacement of 1 inch on oscillogram.

(a) Balance lever (supported at center by (b))
(b) Knife edges.
(c) Circular depressor plate 1-5/8 dia.
(d) Perforated metal base plate 2-1/2 square
(e) Knife edge.
(f) Stabilizing arm (to render the compression surfaces parallel).
(g) Counterbalance.
(h) Pen arm.
(i) Oscillogram (to automatically record deflections)
(j) Micrometer head.
(k) Fastening hook
(l) Test specimen.
(m) Point of application of load (load lever ratio 10 to 1.6)

Dimension in inches.

FIGURE 1 Apparatus for compression deflection test.

4 5.2.2 Procedure The following procedure shall be observed when apparatus shown in Figure 1 is used If an apparatus other than that shown by Figure 1 is used, applicable portions of the following procedure shall apply. The test specimens shall be cut exactly 1 625 inches (4.128 cm) in diameter Specimens taken from either sheets or molded shapes shall have a minimum thickness of 0.25 inch (0.64 cm) and a maximum thickness of 1 125

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inches (2.86 cm). Specimens may have a rind on the top and bottom faces only. All specimens shall have parallel top and bottom faces. The test specimen (1) shall be placed on the base plate (d), and the depressor plate (c) shall be placed on top of the specimen. Several alternate compressions and decompressions shall be made on Type I specimens to expel trapped air. Perfect balance of lever (a) shall be obtained by adjusting the position of counterbalance (g). The lever shall then be fastened to the hook (k), and the micrometer head (j) lowered until it just contacts the depressor plate. A 10-gram preload shall be applied. The hook (k) shall be disengaged, and a reference line traced on the recording drum (i). Standard weights shall be added stepwise at point (m) at 1-minute intervals to deflect the specimen at least 25 percent. Standard weights shall be used which have been so selected that a minimum of four equal weights is required to obtain 25 percent deflection on any cellular test specimen, regardless of its degree of softness or firmness. A load-deflection curve shall be obtained, and the load at 25 percent deflection read from the curve. The load shall be calculated in pounds per square inch (kilopascals). The test shall be run in triplicate and the results averaged. The test specimens shall be used for low temperature resistance (4.5 3).

4.5.3 Low temperature resistance. Low temperature resistance shall be determined as specified in Sections 26 thru 29 of ASTM D 1056 using the test specimens and room temperature load data obtained in 4.5 2 Exposure temperatures shall be as specified in Table VII.

TABLE VII. Low temperature resistance temperatures.

Type	Grade	Temperature
I	A & B	-41° ±1°C (-42° ±2°F)
I	C	-55° ±1°C (-67° ±2°F)
II	A & B	-31° ±1°C (-24° ±2°F)
II	C	-55° ±1°C (-67° ±2°F)

4 5.4 Accelerated aging. Specimens of the dimensions used for testing in 4 5.2.2 shall be aged in a circulating air oven for 70 ±1 hours at 100° ±1°C (212° ±2°F). After aging, the specimens shall be cooled to standard conditions and remain undeflected and undistorted for 22 hours. Each specimen shall then undergo testing as specified in 4.5.2. A load deflection curve shall be obtained by adding 4 equal weights at 1 minute intervals. The total load shall equal that which was required to deflect an unaged specimen to 25 percent (original deflection). The deflection of the aged specimen shall be noted. Percent retention of the original deflection shall be calculated as follows:

$$\text{Retention of original deflection, percent} = W \times 4$$

where W = The deflection of the aged specimen under original deflection load

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4.5.5 Recovery. The thickness of each test specimen (prepared as specified in 4.5.2) shall be measured to 0.001 inch (0.025 mm) and then compressed to 50 percent of original thickness for a period of 46 ±1/2 hours. One minute after removal from the compression apparatus, the thickness of each specimen shall be re-measured and the recovery calculated as follows:

$$\text{Recovery, percent} = \frac{B}{A} \times 100$$

where: A = Original thickness
B = Thickness 1 minute after removal from apparatus.

4.5.6 Flexing. Flexing shall be in accordance with ASTM D 1055 (Sections 24 thru 26) except the test shall be run in triplicate with specimens 2 by 2 by 1 inch (50 by 50 by 25 mm) thick. Each specimen shall undergo 250,000 cycles of flexing The amplitude of compression and decompression shall be 50 percent of the orginal thickness of each specimen. Compression set shall be calculated in accordance with Section 19.1 of ASTM D 1055.

4.5.7 Flame resistance. One end of a strip specimen 0.25 by 0 5 inch (6.35 by 12.7 mm) wide shall be held in a candle (C-C-91, or equal) flame for exactly 1 minute, as shown in Figure 2. The specimen shall be removed from the flame and the time the specimen burns shall be recorded as the flame propagation time. The test shall be performed in an area free from drafts.

4.5.8 Shrinkage (Type II). Shrinkage shall be determined on triplicate specimens 1.625 inches (41.3 mm) in diameter by 1 inch (25.4 mm) thick. The specimens shall be held overnight in a desicator, and then the volume determined by the free-flow seed method, as specified in 4.5.1. After the original volume has been determined, the specimens shall be conditioned in a circulating air oven at 60° ±1°C (140° ±2°F) for 7 days After removal from the oven, the specimens shall be cooled for 2 hours (to room temperature) in a desicator, and again the volume shall be determined by the free-flowing seed method. Percent shrinkage shall be calculated as follows:

$$\text{Percent change in volume} = \frac{V_1 - V_2}{V_1} \times 100$$

where: V_1 = Original volume

V_2 = Volume after oven aging

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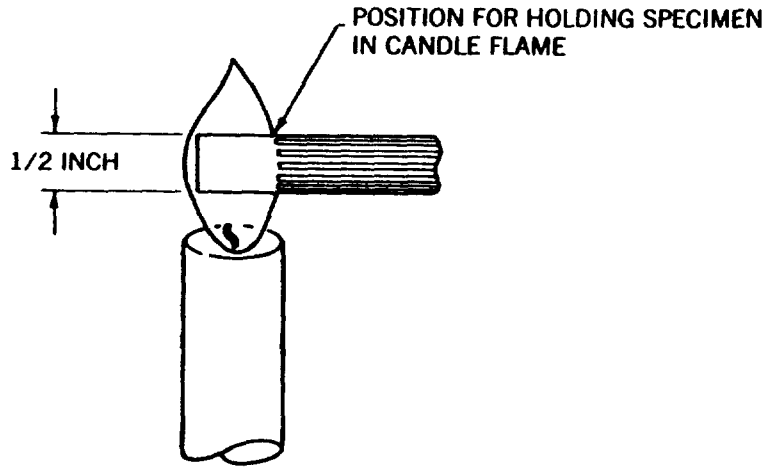


FIGURE 2. Flame-resistance test setup.

4.5.9 Water absorption (Type II). Three test specimen 1.625 inch (41.3 mm) diameter by 1 inch (25.4 mm) thick, shall be individually weighed and then be submerged in a container of distilled water at 24° ±3°C (75° ±5°F) for 22 ±1 hours. The container shall be of sufficient depth to insure that the level of water is 6 inches (15.24 cm) above the top surface of the specimens. Means shall be provided for holding the specimens at the bottom of the containers. After the conditioning period, the specimens shall be dried by compressed air at room temperature until visibly dry and then weighed. The percentage change in weight shall be calculated as follows:

$$\text{Percentage change} = \frac{W_2 - W_1}{W_1} \times 100$$

Where: W_1 = Weight before immersion.

W_2 = Dried weight after immersion

4.5.10 Oil aging. The volume of two test specimens 1.625 inches (41.3 mm) diameter shall be computed by measuring the diameter and thickness to the nearest 0.015 inch (0.38 mm). The specimen shall be immersed in petroleum base reference oil #2 of ASTM D 471 The immersion time shall be 70 hours ±30 minutes at 70° ±1°C (158° ±2°F). At the expiration of the immersion period, the specimen shall be removed and immediately immersed in fresh oil at room temperature for 30 ±5 minutes. The test specimen shall then be rapidly dipped in acetone or alcohol, blotted lightly with filter paper, and measured as above The test shall be run in duplicate and results averaged The percent change in Volume shall then be calculated as follows:

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$$\text{Change in volume (percent)} = \frac{V_1 - V_2}{V_1} \times 100$$

Where: V_1 = Original volume

V_2 = Volume after immersion in oil.

4.5.11 Free sulfur content. The free sulfur content of the material shall be determined in accordance with ASTM D 297.

5. PACKAGING

5.1 Preservation-packaging. Packaging shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. The rubber sheets shall be separated with paper or other suitable separator sheets which will not adhere to or damage the rubber. The molded shapes shall be wrapped, boxed, or otherwise protected against deformation and abrasion.

5.1.2 Commercial. Packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be Level A, B, or Commercial, as specified (see 6.2). Containers shall contain identical amounts of material, be of uniform size and be designed to enclose the contents in a snug, tight-fitting manner. The gross weight of boxes shall not exceed 200 pounds.

5.2.1 Level A. The rubber shall be packed in snug-fitting, overseas, exterior type, boxes conforming to PPP-B-576, PPP-B-591, PPP-B-601, PPP-B-621, or PPP-B-636. Boxes shall be strapped in accordance with the appendix of the applicable box specification.

5.2.2 Level B. Unless otherwise specified, the rubber shall be packed in snug-fitting domestic type boxes conforming to PPP-B-576, PPP-B-591, PPP-B-601, PPP-B-621, or PPP-B-636.

5.2.3 Commercial. The rubber shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking of shipments. In addition to any special markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Type I (open cell). Chemically blown cellular rubber is intended for use in shock absorbing cushions for mounting radio apparatus,

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in crash pads, and other miscellaneous uses requiring a material capable of absorbing shock and damping vibrations It is not recommended for applications involving tensile loading, or where the cellular chemically blown rubber will come in continuous contact with the human skin.

6.1 2 Type II (closed cell). Type II material is intended for flotation gear (skin or rind form only) expansion joint fillers, gaskets, and shock pads where water repellent qualities are desired. It is not recommended for applications where extreme high temperatures are encountered, or where the expansion of the material at high altitudes will be a disadvantage.

6.1.3 Grade. The intended use for grades of Type I (open cell) and for Type II (closed cell) are as follows:

a. Grade A. Grade A cellular rubber is intended for use where resistance to oil and flame is required.

b. Grade B. Grade B cellular rubber is intended for use where resistance to oil is not required.

c. Grade C. Grade C cellular rubber is intended for use where lower-temperature resistance, but not oil resistance, is required.

6.2 Ordering data.

6.2 1 Procurement requirements. Procurement documents should specify the following:

a. Title, number and date of this specification

b. Quantity required.

c. Detail dimensioned drawings or specifications describing the shape, surfacing of the part, dimensions and tolerances.

d. Type, grade and condition of material required (see 1.2).

e Any special characteristics required (odor minimized, special surfacing, etc)

f. Levels of packaging, packing required (Section 5).

g Special markings (5 3).

6 3 Definitions.

6 3.1 Cellular rubber Cellular rubber is defined as a product containing cells or small hollow receptacles. The cells may be either open and interconnected or closed and not interconnected. The cells should be uniform and free from large voids or seams

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6.3.2 Open cell (sponge). Open cell is a rubber-like product made by incorporating into the rubber compound an inflating agent, such as sodium bicarbonate, that gives off a gas which expands the rubber during vulcanization.

6.3.3 Closed cell (expanded). Closed cell is a product usually made by subjecting a rubber compound to a gas, such as nitrogen, under high pressure. It may also be made by incorporating gas-forming materials in the compound.

6.4 Design information. The division into conditions specified herein is based on the assumption that working loads will deflect cellular rubber to not more than 75 percent of its normal or unloaded thickness. Compression deflection, as defined herein, is therefore a measure of the maximum working load which should be applied to any class of material

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army-GL
Navy-AS
Air Force -11

Preparing activity:
Navy-AS

(Project No. 9320-0275)

Review activities:
Army-EA, MR, MI, MR
Navy-OS
Air Force -99
DLA-GS
MISC-DS

User activities:
Army-ME
Navy-MC

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